

In 1965, a Ford Mustang cost \$2,600. While most cars lose value, a 1965 Mustang in immaculate condition can be worth over \$30,000. The function $M(x) = 20x^2 - 420x + 2600$ models the price of the 1965 Mustang in the United States since 1965.

1. What does the variable x represent? Include the units.
2. What does the variable y represent? Include the units.
3. Interpret the y -intercept. Include units.
4. Rewrite the function in vertex form. Show your work.
5. Interpret the vertex in terms of the context. Include units.
6. Complete the table of values for $M(x)$.

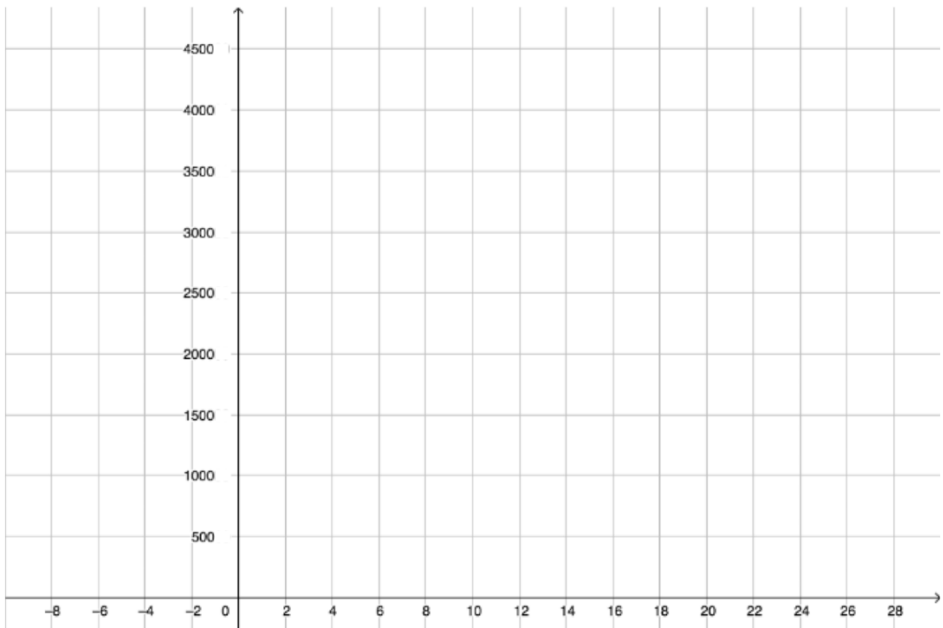
x	-5	-3	0	7	10.5	12	24	50
$M(x)$								

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$			

9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.